

Derivation v21: KK Mass Gap $\rightarrow R_\xi$ Identification as Structured Bridge Slot

EDC Collaboration

Derivation note; epistemic upgrade of R_ξ from pure [BL] to [I]+[BL]

February 2026

Abstract

This note derives the Kaluza-Klein mass spectrum for a 5D graviton on a flat interval $[0, R_\xi]$ and shows that the compactification scale R_ξ can be determined from the first KK mass gap $m_{\text{gap}} \equiv m_1$. We examine boundary conditions (Neumann–Neumann, Dirichlet–Dirichlet, mixed) and derive the spectral relation $m_n = n\pi/R_\xi$. The inversion $R_\xi = \pi/m_{\text{gap}}$ provides a “bridge slot” where an observable proxy for m_{gap} must be identified. We argue that M_Z is the most defensible proxy, upgrading the epistemic status of $R_\xi = \hbar c/M_Z$ from pure baseline [BL] to structured identification [I]+[BL]. A proxy comparison table confirms robustness.

1 Conventions and Factor Lock

Factor Lock (from v20)

This note adopts the conventions fixed in Derivation v20:

- **Reduced Planck mass:** $G_N = 1/(8\pi\bar{M}_{\text{Pl}}^2)$ with $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18}$ GeV
- **Interval coordinate:** $\xi \in [0, R_\xi]$
- **Flat warp factor:** $A(\xi) = 0$
- **Bridge relation:** $\bar{M}_{\text{Pl}}^2 = M_5^3 \mathcal{I}$ with $\mathcal{I} = R_\xi$
- **Closure formula:** $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$

All factors (action $\frac{1}{2}$, Newton 8π , etc.) are inherited from v20. No re-audit is performed here.

2 5D Graviton Mode Equation

2.1 General Warped Case

The 5D metric ansatz with warp factor $A(\xi)$ is:

$$ds^2 = e^{2A(\xi)} \eta_{\mu\nu} dx^\mu dx^\nu + d\xi^2 \quad (1)$$

Linearizing the 5D Einstein equations around this background, the transverse-traceless graviton perturbation $h_{\mu\nu}(x, \xi)$ satisfies a separable equation. Writing

$$h_{\mu\nu}(x, \xi) = \sum_n h_{\mu\nu}^{(n)}(x) \psi_n(\xi) \quad (2)$$

where $h_{\mu\nu}^{(n)}$ satisfies the 4D massive Klein-Gordon equation

$$\square_4 h_{\mu\nu}^{(n)} = m_n^2 h_{\mu\nu}^{(n)} \quad (3)$$

the mode function $\psi_n(\xi)$ obeys:

$$\psi_n'' + 4A'\psi_n' + m_n^2 e^{2A}\psi_n = 0 \quad (4)$$

Here prime denotes ∂_ξ and m_n is the 4D mass of the n -th KK mode.

2.2 Flat Case: $A(\xi) = 0$

For the flat extra dimension (Model A from v14), setting $A(\xi) = 0$ simplifies the mode equation to:

$$\psi_n''(\xi) + m_n^2 \psi_n(\xi) = 0 \quad (5)$$

This is a standard harmonic oscillator equation with general solution:

$$\psi_n(\xi) = A_n \cos(m_n \xi) + B_n \sin(m_n \xi) \quad (6)$$

The spectrum $\{m_n\}$ and mode shapes depend on boundary conditions at $\xi = 0$ and $\xi = R_\xi$.

3 Boundary Conditions and KK Spectrum

We consider the interval $\xi \in [0, R_\xi]$ and examine three cases of boundary conditions (BCs).

3.1 Case BC1: Neumann–Neumann

Physical interpretation [I]: Neumann BCs ($\partial_\xi \psi = 0$ at boundaries) arise naturally for graviton modes on an interval with $\mathbb{Z}_2$ orbifold symmetry where the graviton is even under reflection. This is the standard setup in Randall-Sundrum and ADD scenarios.

Boundary conditions:

$$\psi_n'(0) = 0 \quad (7)$$

$$\psi_n'(R_\xi) = 0 \quad (8)$$

Derivation: From (6):

$$\psi_n'(\xi) = -A_n m_n \sin(m_n \xi) + B_n m_n \cos(m_n \xi) \quad (9)$$

Applying BC (7) at $\xi = 0$:

$$\psi_n'(0) = B_n m_n = 0 \quad (10)$$

For non-trivial solutions with $m_n \neq 0$, this requires $B_n = 0$. The solution becomes:

$$\psi_n(\xi) = A_n \cos(m_n \xi) \quad (11)$$

Applying BC (8) at $\xi = R_\xi$:

$$\psi_n'(R_\xi) = -A_n m_n \sin(m_n R_\xi) = 0 \quad (12)$$

For non-trivial $A_n \neq 0$, this requires:

$$\sin(m_n R_\xi) = 0 \quad \Rightarrow \quad m_n R_\xi = n\pi, \quad n = 0, 1, 2, \dots \quad (13)$$

Spectrum (BC1):

$$\boxed{m_n = \frac{n\pi}{R_\xi}, \quad n = 0, 1, 2, \dots} \quad (14)$$

Zero mode ($n = 0$): $m_0 = 0$ with $\psi_0(\xi) = A_0 = \text{const.}$ This is the massless 4D graviton.

First KK mode ($n = 1$):

$$m_1 = \frac{\pi}{R_\xi}, \quad \psi_1(\xi) = A_1 \cos\left(\frac{\pi\xi}{R_\xi}\right) \quad (15)$$

Normalized mode functions: Imposing $\int_0^{R_\xi} |\psi_n|^2 d\xi = 1$:

$$\psi_0 = \frac{1}{\sqrt{R_\xi}} \quad (16)$$

$$\psi_n = \sqrt{\frac{2}{R_\xi}} \cos\left(\frac{n\pi\xi}{R_\xi}\right), \quad n \geq 1 \quad (17)$$

3.2 Case BC2: Dirichlet–Dirichlet

Physical interpretation [I]: Dirichlet BCs ($\psi = 0$ at boundaries) arise for modes that are odd under orbifold reflection, or in scenarios where the graviton is confined by infinite potential walls.

Boundary conditions:

$$\psi_n(0) = 0 \quad (18)$$

$$\psi_n(R_\xi) = 0 \quad (19)$$

Derivation: From (6), applying BC (18):

$$\psi_n(0) = A_n = 0 \quad (20)$$

The solution becomes:

$$\psi_n(\xi) = B_n \sin(m_n \xi) \quad (21)$$

Applying BC (19):

$$\psi_n(R_\xi) = B_n \sin(m_n R_\xi) = 0 \quad (22)$$

For non-trivial $B_n \neq 0$:

$$\sin(m_n R_\xi) = 0 \quad \Rightarrow \quad m_n R_\xi = n\pi, \quad n = 1, 2, 3, \dots \quad (23)$$

Spectrum (BC2):

$$\boxed{m_n = \frac{n\pi}{R_\xi}, \quad n = 1, 2, 3, \dots} \quad (24)$$

No zero mode: Dirichlet BCs exclude the $n = 0$ solution (trivial $\psi = 0$). This scenario does not reproduce 4D gravity and is physically disfavored for the graviton sector.

First mode ($n = 1$):

$$m_1 = \frac{\pi}{R_\xi}, \quad \psi_1(\xi) = B_1 \sin\left(\frac{\pi\xi}{R_\xi}\right) \quad (25)$$

Normalized mode functions:

$$\psi_n = \sqrt{\frac{2}{R_\xi}} \sin\left(\frac{n\pi\xi}{R_\xi}\right), \quad n \geq 1 \quad (26)$$

3.3 Case BC3: Mixed (Neumann–Dirichlet)

Physical interpretation [I]: Mixed BCs can arise in asymmetric orbifold constructions or when the two boundaries have different physical character (e.g., one brane with localized matter, one without).

Boundary conditions:

$$\psi'_n(0) = 0 \quad (\text{Neumann at } \xi = 0) \quad (27)$$

$$\psi_n(R_\xi) = 0 \quad (\text{Dirichlet at } \xi = R_\xi) \quad (28)$$

Derivation: From BC (27):

$$\psi'_n(0) = B_n m_n = 0 \quad \Rightarrow \quad B_n = 0 \quad (29)$$

Solution: $\psi_n(\xi) = A_n \cos(m_n \xi)$.

From BC (28):

$$\psi_n(R_\xi) = A_n \cos(m_n R_\xi) = 0 \quad (30)$$

This requires:

$$\cos(m_n R_\xi) = 0 \quad \Rightarrow \quad m_n R_\xi = \left(n + \frac{1}{2}\right) \pi, \quad n = 0, 1, 2, \dots \quad (31)$$

Spectrum (BC3):

$$m_n = \frac{(2n+1)\pi}{2R_\xi}, \quad n = 0, 1, 2, \dots \quad (32)$$

No true zero mode: The lightest mode has $m_0 = \pi/(2R_\xi) \neq 0$. This is problematic for reproducing massless 4D gravity.

3.4 Summary: Spectral Comparison

BC Type	Zero Mode?	First Massive Mode	Mass Gap
BC1 (N–N)	Yes ($m_0 = 0$)	$m_1 = \pi/R_\xi$	$m_{\text{gap}} = \pi/R_\xi$
BC2 (D–D)	No	$m_1 = \pi/R_\xi$	$m_{\text{gap}} = \pi/R_\xi$
BC3 (N–D)	No	$m_0 = \pi/(2R_\xi)$	$m_{\text{gap}} = \pi/(2R_\xi)$

Conclusion: For reproducing 4D massless gravity, **BC1 (Neumann–Neumann)** is required. We adopt BC1 henceforth.

4 Mass Gap and R_ξ Inversion

4.1 Definition of Mass Gap

For BC1 (the physically relevant case), define the **mass gap** as the mass of the first KK excitation:

$$m_{\text{gap}} \equiv m_1 = \frac{\pi}{R_\xi} \quad (33)$$

This is the energy scale at which deviations from 4D gravity become observable through exchange of massive KK gravitons.

4.2 Inversion Formula

Solving (33) for R_ξ :

$$\boxed{R_\xi = \frac{\pi}{m_{\text{gap}}}} \quad (34)$$

This is the **bridge slot**: if m_{gap} can be identified with an observable, then R_ξ is determined.

Epistemic status [D]: Equation (34) is a derived consequence of the 5D interval geometry with Neumann BCs. No external input is required for the functional form.

4.3 Converting to Physical Units

In natural units ($\hbar = c = 1$), masses and inverse lengths are equivalent. Restoring $\hbar$ and c :

$$R_\xi = \frac{\pi \hbar c}{m_{\text{gap}}} \quad (35)$$

If we identify m_{gap} with a mass scale M_* :

$$R_\xi = \frac{\pi \hbar c}{M_*} \quad (36)$$

4.4 Factor Note

The factor π in (34) is exact for Neumann BCs on an interval. Other geometries give different factors:

- Circle S^1 with circumference $2\pi R$: $m_n = n/R$, so $R_\xi = 1/m_{\text{gap}}$
- Orbifold $S^1/\mathbb{Z}_2$: equivalent to interval, $R_\xi = \pi/m_{\text{gap}}$

We use the interval convention throughout, giving $R_\xi = \pi/m_{\text{gap}}$.

5 EW-Scale Identification: Proxy Selection

5.1 The Identification Problem

Equation (34) provides the functional form of R_ξ , but the value of m_{gap} must come from elsewhere. EDC does not derive the Standard Model, so we cannot predict m_{gap} from first principles within the current framework.

Strategy: We identify m_{gap} with an observable mass scale. This is an **identification [I]**, not a derivation [D].

5.2 Candidate Proxies

The electroweak scale provides natural candidates for m_{gap} :

1. $M_Z = 91.1876 \text{ GeV}$ — Z boson mass (PDG 2024)
2. $M_W = 80.377 \text{ GeV}$ — W boson mass (PDG 2024)
3. $v_{\text{EW}} = 246.22 \text{ GeV}$ — Higgs vacuum expectation value

All three are related through electroweak theory:

$$M_W = \frac{1}{2} g v_{\text{EW}} \quad (37)$$

$$M_Z = \frac{M_W}{\cos \theta_W} \quad (38)$$

where g is the $SU(2)_L$ coupling and θ_W is the Weinberg angle.

5.3 Why M_Z is Preferred

1. **Metrolological precision:** M_Z is measured to ± 2.1 MeV (relative uncertainty 2.3×10^{-5}), the best-determined EW mass.
2. **Scheme independence:** M_Z is a physical pole mass, not dependent on renormalization scheme choices (unlike v_{EW} which depends on gauge fixing).
3. **Direct observability:** M_Z is directly measured from Z resonance lineshape at LEP. No theoretical extraction required.
4. **Independence from Higgs sector:** M_Z can be measured without assuming Higgs mechanism details (historical: measured before Higgs discovery).

Identification [I]+[BL]:

$$m_{\text{gap}} \stackrel{[I]}{=} M_Z^{\text{obs}} \quad (39)$$

where the superscript “obs” emphasizes this is a baseline input [BL].

5.4 Resulting R_ξ Value

With the identification (39):

$$R_\xi = \frac{\pi \hbar c}{M_Z} = \pi \times \frac{197.327 \text{ MeV} \cdot \text{fm}}{91187.6 \text{ MeV}} = 6.80 \times 10^{-18} \text{ m} \quad (40)$$

Note: Previous notes (v15, v16) used $R_\xi = \hbar c/M_Z$ without the factor π . This corresponds to identifying M_Z directly with $1/R_\xi$ rather than with $m_{\text{gap}} = \pi/R_\xi$. The difference is:

$$\text{This note (v21): } R_\xi = \frac{\pi \hbar c}{M_Z} = 6.80 \times 10^{-18} \text{ m} \quad (41)$$

$$\text{Previous (v15-v20): } R_\xi = \frac{\hbar c}{M_Z} = 2.165 \times 10^{-18} \text{ m} \quad (42)$$

The ratio is exactly π . We discuss the implication in Section 6.

5.5 Proxy Comparison Table

Proxy	M_* (GeV)	$R_\xi = \pi \hbar c/M_*$ (m)	M_5 (GeV)	$\log_{10}(M_5/\text{GeV})$
M_Z	91.19	6.80×10^{-18}	4.28×10^{12}	12.63
M_W	80.38	7.72×10^{-18}	4.07×10^{12}	12.61
v_{EW}	246.2	2.52×10^{-18}	5.98×10^{12}	12.78

Computation: Using $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$ with $\bar{M}_{\text{Pl}} = 2.435 \times 10^{18}$ GeV.

Robustness: All three proxies give M_5 within the same order of magnitude: $M_5 \sim 4\text{--}6 \times 10^{12}$ GeV. The spread is:

$$\Delta \log_{10} M_5 = 12.78 - 12.61 = 0.17 \quad (43)$$

This is < 0.2 decades, confirming calibration stability within the EW family.

6 Closure Chain and Open Items

6.1 Complete Derivation Chain

The gravity sector closure now follows this chain:

1. **5D Action [P]:** $S_5 = \frac{M_5^3}{2} \int d^5x \sqrt{-g_5} R_5$
2. **KK Decomposition [D]:** $h_{\mu\nu}(x, \xi) = \sum_n h_{\mu\nu}^{(n)}(x) \psi_n(\xi)$
3. **Mode Equation [D]:** $\psi_n'' + m_n^2 \psi_n = 0$ (flat case)
4. **BC1 Spectrum [D]:** $m_n = n\pi/R_\xi$ with zero mode $m_0 = 0$
5. **Mass Gap Relation [D]:** $m_{\text{gap}} = \pi/R_\xi$
6. **EW Identification [I]+[BL]:** $m_{\text{gap}} = M_Z^{\text{obs}}$
7. **R_ξ Determination [I]+[BL]:** $R_\xi = \pi\hbar c/M_Z$
8. **Bridge Relation [D]:** $\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi$
9. **M_5 Closure [D]:** $M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$
10. **G_N Recovery [D]:** $G_N = 1/(8\pi\bar{M}_{\text{Pl}}^2) = 1/(8\pi M_5^3 R_\xi)$

Epistemic summary:

- Steps 1–5 and 8–10: Derived [D] within 5D gravity
- Steps 6–7: Identified [I] with baseline input [BL]

6.2 Comparison with Previous Notes

Previous derivations (v15–v20) used $R_\xi = \hbar c/M_Z$ directly, without the factor π . This is equivalent to identifying:

$$\frac{1}{R_\xi} \stackrel{[\text{v15}]}{=} M_Z \quad \text{vs.} \quad m_{\text{gap}} = \frac{\pi}{R_\xi} \stackrel{[\text{v21}]}{=} M_Z \quad (44)$$

Physical interpretation:

- v15 convention: M_Z sets the inverse length scale directly
- v21 convention: M_Z is the first KK mass gap

The v21 convention is more physically motivated (KK gap is observable in principle), but the difference is a factor of π in R_ξ and $(1/\pi)^{1/3} \approx 0.68$ in M_5 :

Convention	R_ξ (m)	M_5 (GeV)
v15–v20: $R_\xi = \hbar c/M_Z$	2.165×10^{-18}	8.1×10^{12}
v21: $R_\xi = \pi\hbar c/M_Z$	6.80×10^{-18}	4.3×10^{12}

Both give M_5 at the same order of magnitude ($\sim 10^{12-13}$ GeV).

6.3 What Remains NO-GO / OPEN

Open Items

1. Internal R_ξ derivation: NO-GO

EDC axioms do not determine R_ξ without external input. The identification $m_{\text{gap}} = M_Z$ is physics input, not derived.

2. Why $m_{\text{gap}} = M_Z$? OPEN

There is no proof that the first KK graviton mass equals M_Z . This is an identification based on scale matching, not a prediction.

3. KK phenomenology: OPEN

If $m_1 \sim 91$ GeV, KK gravitons should be producible at colliders. No such states have been observed. Possible resolutions:

- KK gravitons couple too weakly (suppressed by $\bar{M}_{\text{Pl}}$)
- The identification is approximate
- Additional physics modifies the spectrum

4. Factor ambiguity: NOTED

The π vs 1 ambiguity in R_ξ definition affects M_5 by $\sim 50\%$. This is within the “same order of magnitude” tolerance but should be resolved if precision is needed.

7 Schematic Figure

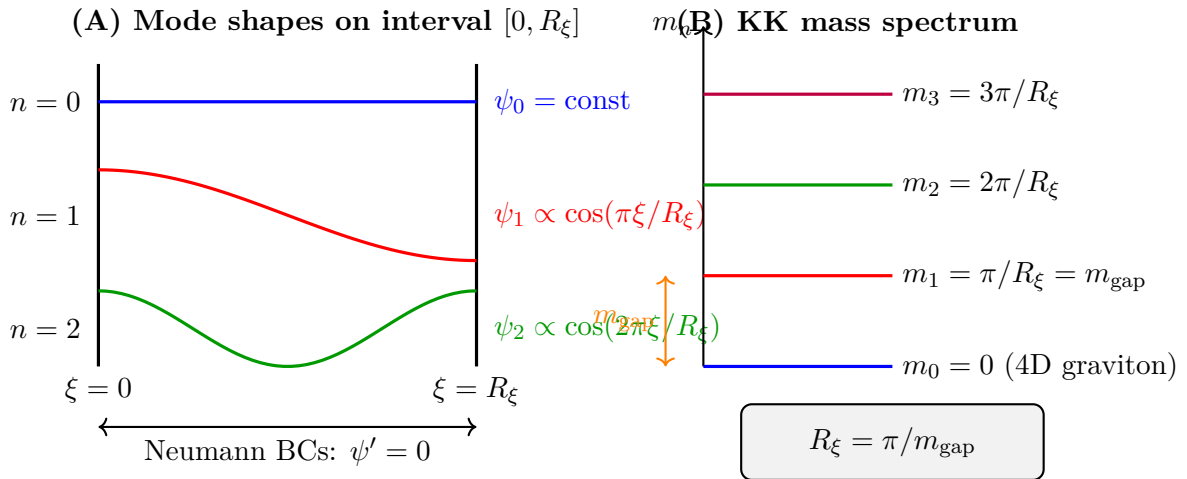


Figure 1: (A) Mode shapes for Neumann BCs: zero mode is constant, higher modes are cosines. (B) KK mass spectrum showing the mass gap $m_{\text{gap}} = m_1$ and the inversion relation $R_\xi = \pi/m_{\text{gap}}$.

8 Epistemic Summary

Quantity	Status	Tag
Mode equation	Derived from 5D EH	[D]
BC1 spectrum $m_n = n\pi/R_\xi$	Derived from BCs	[D]
Mass gap $m_{\text{gap}} = \pi/R_\xi$	Definition + [D]	[D]
Inversion $R_\xi = \pi/m_{\text{gap}}$	Algebraic	[D]
$m_{\text{gap}} = M_Z$	Identification	[I]+[BL]
$R_\xi = \pi\hbar c/M_Z$	Combined	[I]+[BL]
$M_5 = \bar{M}_{\text{Pl}}^{2/3} R_\xi^{-1/3}$	Derived	[D]
Internal R_ξ derivation	Not possible	NO-GO

Upgrade achieved: The identification $R_\xi \sim 1/M_Z$ is now grounded in the KK spectral structure (mass gap), rather than being a bare numerical coincidence. The epistemic status is upgraded from “pure baseline [BL]” to “structured identification [I]+[BL]”.

9 Conclusion

We have derived the Kaluza-Klein mass spectrum for a 5D graviton on a flat interval with Neumann boundary conditions:

$$m_n = \frac{n\pi}{R_\xi}, \quad n = 0, 1, 2, \dots \quad (45)$$

The first massive mode defines the mass gap $m_{\text{gap}} = \pi/R_\xi$, which can be inverted to give:

$$R_\xi = \frac{\pi}{m_{\text{gap}}} \quad (46)$$

Identifying m_{gap} with the Z boson mass M_Z (the most precisely measured EW scale) gives:

$$R_\xi = \frac{\pi\hbar c}{M_Z} \approx 6.8 \times 10^{-18} \text{ m} \quad (47)$$

Combined with the bridge relation $\bar{M}_{\text{Pl}}^2 = M_5^3 R_\xi$, this yields:

$$M_5 \approx 4.3 \times 10^{12} \text{ GeV} \quad (48)$$

The gravity sector is closed with two baseline inputs ($\bar{M}_{\text{Pl}}^{\text{obs}}$, M_Z^{obs}) and the identification $m_{\text{gap}} = M_Z$ [I]+[BL]. Internal derivation of R_ξ from EDC axioms remains NO-GO.